Displayed Prices, Hidden Clearing: Property Tests and Mechanism Design for AI Inference Routing

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Abstract

Open-weight models create a real-time market for partly substitutable execution. Harnesses originate token demand, routers choose where to buy execution, and providers sell perishable compute under a common model label. We ask what can be learned about this market when prices are public but clearing, capacity, and quality are only partly observed. Five-minute menus, router enforcement state, and owned requests reject a simple public-order-book description: a documented inverse-price heuristic is not the realized allocation law, active undercutting is transient, and a registered response screen finds no excess rival reaction. We then derive mechanism results that survive this limited identification. A price-only Gibbs router has an exact welfare-regret identity for omitted service attributes; uniformly accurate scores are near-optimal; but no quote-only rule can be robustly efficient when reserved capacity is hidden. These results imply an auditable design using execution-contingent prices, capacity commitments, quality scores, exploration, and fees separated from selected spend. A held-out replay reduces a quote-coupling proxy by 49.3% at a 2.08% displayed- price premium, but does not identify welfare. The contribution is a set of property tests and implementable mechanism bounds, not a conduct claim.

1 Introduction

Open weights separate a model from the firm that executes it. Two providers can serve the same model on the same prompt, making execution partly commodity-like. Yet tool calls, quantization, sampling, context limits, latency, privacy, and hidden capacity keep the products from being perfectly fungible. This produces a new vertical market: a user delegates to a harness; the harness chooses a model and possibly a router; the router aggregates token demand and dispatches an eligible provider; and inference providers sell perishable GPU-backed service.

The closest DeFi analogy is useful but incomplete. A harness resembles a wallet or intent originator, a router resembles a DEX aggregator, a model is a common execution standard, and a provider resembles a liquidity provider. Unlike an atomic swap, however, an inference request is normally assigned to a single provider, may be retried serially, has stochastic quality, and cannot be verified from price alone. The router is therefore not merely an aggregator. It is a delegated buyer, a multi-attribute scoring mechanism, and a reliability intermediary.

This distinction matters empirically. A public price is an invitation to route, not a firm capacity commitment. A documented score is not necessarily the allocation law after private eligibility, quality, and fallback. Provider cost depends on whether capacity is owned, reserved, or bought in a spot market; the same discount can reflect low marginal cost, subsidy, entry investment, or predatory intent. Consequently, price panels alone neither reveal flow nor identify dumping, collusion, or welfare.

We build the paper around this identification problem. Our evidence joins a five-minute public menu panel, router-published enforcement aggregates, model- author anchor prices, public quality fields, and randomized or assignment-first owned requests. The sources occupy different rungs of an identification ladder. Public menus reveal offers; public enforcement reveals router-side state; mechanical shadow routing reveals the implications of an explicit rule; owned requests reveal realized policy for one account and workload; and only router or provider logs could reveal market-wide flow or private capacity.

The empirical results are primarily property tests. They tell us which simple market descriptions are incompatible with the observations, and thereby narrow the set of mechanisms worth designing. Four posted-price regimes are highly persistent except for active undercutting. A registered response screen finds less rival reaction after active cuts than in a shifted timing placebo. The public inverse-square heuristic is rejected as realized allocation by an independent elasticity audit. Premium quotes do not have a detectable quality- of-service advantage in the current holdout. Owned routing remains too narrow to infer market share. An HMP-style signal-coupling screen has the predicted sign but fails its randomization test, concentration gate, and downstream power gates. These results do not say strategic interaction is absent; they say its current identified set is large.

We make three mechanism-design contributions.

(1) We formulate separate user, router, provider, and planner objectives for inference routing. The formulation makes explicit why price-only, router- revenue, and welfare objectives need not agree.
(2) For entropy-regularized routing, we prove an exact regret identity: using a price-only score instead of the full generalized execution cost loses exactly a scaled KL divergence. A uniform score-error bound then gives a robust approximation guarantee.
(3) We prove a hidden-capacity impossibility result. Two economies with the same public menu can require opposite efficient allocations. Every quote-only router incurs at least half the cost gap in one of them. Execution-contingent capacity certificates are therefore an informational requirement, not merely a feature request.

Finally, we evaluate an implementable adaptive score in held-out historical menus and a strategic multi-agent simulator. The replay exposes a real trade- off between a coupling proxy and displayed price but holds provider behavior fixed. A marginal-preserving signal-order intervention raises buyer prices in a focal two-UCB environment but vanishes across heterogeneous learning rules. These are useful design screens, not live-market welfare estimates.

The paper contributes to multidimensional procurement [3], two-sided platforms [14], pricing technology [2], platform steering [9], and algorithmic interaction [6]. Its methodological position is deliberately conservative: when market-wide flow and cost are hidden, theory should state conditional guarantees and experiments should test mechanism properties before assigning conduct.

2 Market, Timing, and Objectives

2.1 Participants and transaction

A request has observable context x and user value $v(x, q)$ for delivered quality q. A harness transforms the task and chooses a model-router pair. The router observes a private eligibility set $A(x)$, public or private endpoint features, and possibly an ordered fallback list. Provider i posts input and output token prices, summarized by an expected request price $p_i(x)$, and delivers stochastic latency L_i, quality q_i, and success $y_i \in \{0, 1\}$.

Providers differ in capacity technology. An owned-capacity provider chooses K_i before demand, pays fixed cost $F_i(K_i)$, and may have low short-run marginal cost. A spot-dependent provider buys state-contingent capacity at $c_i(x, z)$ and has less fixed exposure. A model author can post a focal anchor price. A provider matching that anchor is "non-strategic" only with respect to its displayed-price process; it can still optimize admission, quality, or capacity.

Let s_i be the probability that i is attempted, including any position in a declared fallback policy, and let $d_i = s_i \mathbb{E}[y_i \mid i, x]$ be delivered share. Transfers cancel in static social surplus but affect entry and investment. The four objectives are

$$U_i = \mathbb{E}[(p_i - c_i(x, K_i, q_i))d_i - F_i(K_i) - \phi_i], \tag{1}$$

$$R = \mathbb{E}[f(p_i, d_i) - C_{\text{retry}} - C_{\text{SLA}}], \tag{2}$$

$$V = \mathbb{E}[v(x, q_i)y_i - p_i y_i - \lambda_L L_i - \lambda_F(1 - y_i)], \tag{3}$$

$$W = V + R + \sum_i U_i - \text{pure transfers}, \tag{4}$$

where ϕ_i includes penalties or collateral losses. A router may maximize revenue R, a user-facing generalized cost, or a constrained combination. We never infer which objective is deployed from public prices.

2.2 Information ladder

Table 1 states what each source can identify. The distinction between simulated share and realized share is essential. If an analyst applies $p_i^{-2}/\sum_j p_j^{-2}$ to a public menu, the output is a property of that formula and menu. It is not routed share unless eligibility, quality filters, and fallback are either observed or experimentally shown irrelevant.

Table 1: Identification ladder. Higher rows do not inherit the claims of lower rows without an explicit bridge.

Source	Identified object
Public menu	Quotes, endpoint labels, displayed availability
Router aggregates	Published congestion, derank, and rate-limit state
Shadow rule	Allocation implied by a declared formula, conditional on menu
Owned requests	Selected provider, cost, and outcomes for our account/workload
Private logs	Potentially arrivals, eligibility, attempts, and market-wide flow
Cost/value data	Needed with outcomes for welfare and dumping claims

3 Conditional Mechanism Results

3.1 Inverse-power routing is a benchmark, not a measurement

Suppose, only for this subsection, that all n providers are eligible and the router really allocates

$$s_i(p) = \frac{p_i^{-a}}{\sum_j p_j^{-a}}, \qquad a > 0. \tag{5}$$

With constant marginal cost c_i and fixed demand, provider profit is $\pi_i = (p_i - c_i)s_i$. This is a useful mechanism benchmark even though our data reject it as the full realized allocation law.

Proposition 1 (Conditional share elasticity and markup). *Under* (5),

$$\frac{\partial \log s_i}{\partial \log p_i} = -a(1 - s_i). \tag{6}$$

Every interior best response satisfies the Lerner index $L_i = (p_i - c_i)/p_i = 1/[a(1 - s_i)] > 1/a$. If costs are symmetric, an interior symmetric equilibrium exists when $a(n - 1) > n$ and has

$$p^* = c\frac{a(n-1)}{a(n-1) - n}. \tag{7}$$

The result explains why the routing exponent controls price incentives *when the mechanism is actually used.* It does not license substituting $a = 2$ into a structural market model. Private eligibility and quality change both s_i and the derivative providers face.

3.2 The cost of omitting quality

Let the generalized execution cost of provider i be

$$g_i = a \log p_i + \lambda_L \widehat{L}_i + lambda_F \widehat{r}_i - \lambda_Q \widehat{q}_i - \lambda_K \log \widehat{K}_i, \tag{8}$$

where $\widehat{r}_i$ is failure risk and $\widehat{K}_i$ is committed, not merely advertised, capacity. For x in the probability simplex Δ_n, define the entropy-regularized objective

$$F_g(x) = \langle x, g \rangle + \eta^{-1} \sum_i x_i \log x_i, \qquad \eta > 0. \tag{9}$$

Its minimizer is $x_i^g \propto e^{-\eta g_i}$. A price-only router instead uses $\tilde{g}_i = a \log p_i$ and $x^{\tilde{g}}$.

THEOREM 1 (EXACT OMITTED-ATTRIBUTE REGRET). *For any finite score vector g and any allocation $x \in \Delta_n$,*

$$F_g(x) - F_g(x^g) = \eta^{-1}\mathrm{KL}(x \,\|\, x^g). \tag{10}$$

In particular, the loss from price-only routing is exactly $\eta^{-1}\mathrm{KL}(x^{\tilde{g}} \| x^g)$ and is zero if and only if the two allocations coincide. Omitted attributes that are constant across eligible providers do no harm; attributes that reorder providers generally do.

The identity is sharper than saying quality "may matter." It turns the distance between a price-only and full score into an exact objective gap for a declared regularized objective. It also gives a measurable target: a router can publish both score components and report the implied KL gap without disclosing prompt contents.

THEOREM 2 (ROBUST APPROXIMATE SCORING). *Suppose $\|\widehat{g}-g\|_\infty \le \epsilon$ and let $x^{\widehat{g}}$ minimize $F_{\widehat{g}}$. Then*

$$0 \le F_g(x^{\widehat{g}}) - F_g(x^g) \le 2\epsilon. \tag{11}$$

Thus an auditable score can be approximately efficient without perfectly estimating every attribute. The guarantee is pointwise and does not assume a provider behavioral model. It does require that the score target g encode the planner or user objective rather than an ad hoc proxy.

3.3 Why public prices cannot recover hidden capacity

THEOREM 3 (QUOTE-ONLY HIDDEN-CAPACITY IMPOSSIBILITY). *Consider two eligible providers with identical observable quotes, public quality, and public reliability. There exist two economies consistent with those observables: in economy A, provider 1 has committed capacity and social cost c while provider 2 has scarcity cost $c + \Delta$; in economy B the costs are reversed. Any quote-only randomized router incurs expected cost regret at least $\Delta/2$ in one economy. A deterministic quote-only router incurs regret Δ in one economy.*

The theorem formalizes the reserved-capacity problem. Fundraising, data-center ownership, or a provider name is not the missing statistic: those are noisy proxies for deliverable capacity. The mechanism needs a commitment whose breach has a consequence. A capacity certificate can be a finite-slot report backed by collateral, an execution-contingent price, or an SLA with measurable credits.

3.4 Fee conflict

Suppose the router receives an ad-valorem fee τ on selected spend. Holding volume and allocation fixed, its revenue is $\tau \sum_i x_i p_i$, while the user's payment is $\sum_i x_i p_i$. A common price increase therefore benefits the router and harms the user one-for-one up to τ. A flat fee per successful request is locally neutral to the selected price but can still distort admission or volume. Fee separation does not solve mechanism design, but it removes the most direct price-level conflict.

4 Data and Property Tests

4.1 Capture system and estimands

The main public panel begins on 7 July 2026 and captures endpoint and model state at five-minute cadence in a remote GitHub Actions–Hugging Face pipeline. We construct completion-price events only when a same endpoint-model quote changes between valid snapshots. We separately collect router-published rate-limit, derank, and capacity fields. Assignment-first paid probes record treatment before sending a minimal request and retrieve the selected provider and outcome for our account. Private prompts and request rows never enter public releases.

Every result freezes a dataset revision and declares a calibration/holdout split. The provider-regime analysis uses revision `cc045919595d2f8b0c211c128ac3e535f9b8c008`. The first HMP monitor uses `aeb7f1656698885145481706O6fd3a29b1d0fb5f`. The adaptive replay is trained before 16 July and evaluated afterward. These windows are short; they support falsification and mechanism screening, not a mature structural estimate.

4.2 Regimes are persistent, except active undercutting

Provider-model pairs are frozen in a training window into four descriptive pricing regimes relative to the model-author benchmark: premium, anchor adopter, static discounter, and active undercutter. The names describe a price path, not a provider's cost or intent. Of 365 training pairs, 360 are holdout-evaluable. Overall persistence is 0.892 (95% Wilson interval $[0.855, 0.920]$). Premium and static-discount regimes persist at 0.991 and 0.962; anchor adoption at 0.830; but active undercutting persists in only 4 of 20 pairs, or 0.200 ($[0.081, 0.416]$). Fourteen of the 20 active pairs become static discounters.

This pattern fits a Brown–MacKay-style world in which repricing technologies differ [2], but it does not identify such technology. Active undercutting could also reflect promotion windows, capacity shocks, or menu corrections. The testable implication is duration: future data should estimate transition hazards as a function of benchmark gap, capacity state, and update frequency, with provider-model fixed effects.

4.3 No excess rival response in the active regime

For 1,006 active-undercutter events in 11 target provider-model clusters, the registered rival-action rate is 0.376 ($[0.346, 0.406]$), compared with 0.456 ($[0.426, 0.487]$) in a shifted timing placebo. The paired difference is -0.081; a target-cluster bootstrap interval is $[-0.132, 0]$, and the exact one-sided test for *more* registered-window response has $p = 1.00$. The placebo is not randomized treatment, but the direction rejects a simple "undercut then immediate rival response" account in the current window.

4.4 Displayed score is not realized clearing

An independent welfare audit compares the elasticity implied by the public inverse-square score with the observed aggregate share elasticity. The inverse-square restriction is rejected with $z = 10.06$. An earlier aggregate estimate is -1.19 (standard error 0.17), but it is neither request-level selection nor a price experiment. We therefore

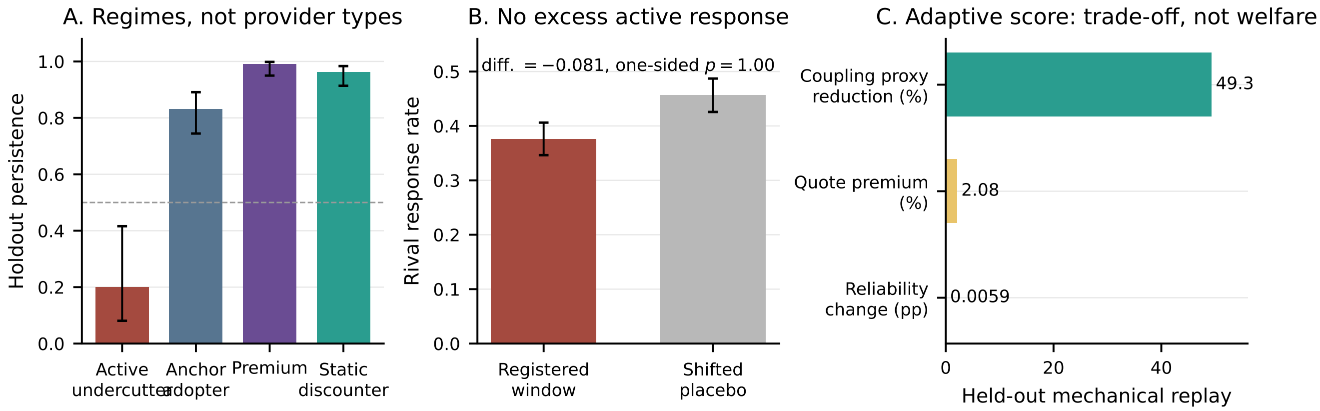

Figure 1: Three empirical boundaries. Panel A: frozen training regimes and holdout persistence (Wilson 95% intervals). Panel B: active-undercutter rival response in the registered window versus a shifted timing placebo. Panel C: held-out adaptive-score replay; the reliability change is in percentage points. Panel C holds menus and provider behavior fixed, so it is not a welfare estimate.

use inverse-power routing only as the conditional benchmark in Section 3.

Owned routing reinforces the distinction. In the provider-regime release, 151 delegated selections match frozen regimes: 70 anchor, 73 premium, four active, and four static. These counts depend on our model and prompt mix and are not market shares. A separate frozen first-position experiment, H81, decomposes fallback from delegated selection. In its 133 preterminal assignments, the missing-outcome identified set places the fallback and total-delegation contrast at 4.4–6.7 percentage points and hidden selection at zero. Missingness suppresses the complete-data primary randomization family, and two repeated models prevent broad transport. The fixed-horizon H95 replication remains blinded until its preregistered horizon and is not pooled with H81.

4.5 Quality, capacity, and "dumping" remain unidentified

Premium providers do not show a significant holdout throughput advantage ($p = 0.392$) or latency advantage ($p = 0.738$) in provider-model residual aggregates. This does not prove premiums lack value: task fidelity, privacy, support, and selection into quality measurement remain missing. More sharply, zero provider-model cells have a valid utilization measure linking load to a capacity ceiling. Static discounters more often fall below one scenario GPU- cost bound, but marginal cost, reserved-capacity amortization, rebates, and subsidies are unobserved. The registered verdict is therefore `dumping_supported=false`.

The same caution applies to geography. A name-based "China-linked" tag is more common among active and static discount regimes, but a name tag is neither cost nor strategy. A causal explanation requires at least location, hardware, energy price, reserved capacity, subsidy/rebate exposure, and realized utilization. These are fields for a data partnership, not coefficients to impute from names.

5 Signal Coupling and Collusion Screens

Hansen–Misra–Pai show how independent pricing algorithms can reach high prices when correlated experiments bias own-price learning [6]. Their mechanism does not require communication. It does require a chain of observable properties before it can explain this market:

(1) excess residual quote coupling after preserving common clocks;
(2) a preperiod routing signal-to-noise gradient;
(3) a wedge between independent and coupled elasticity specifications; and
(4) a forward transition toward higher buyer prices or high-price states.

WF18 freezes this chain and a Holm family. In its first monitor, 686 price events span 44 models and ten providers. The mechanism window has 253 events and 19 provider-pair/model clusters. The 24-hour residual covariance is 0.01659; its excess over 2,000 clock-preserving circular shifts is 0.01874, with a one-sided $p = 0.10995$ and null 95% interval $[-0.03056, 0.02444]$. One provider pair supplies 76.7% of paired events, and leave-one-unit estimates range from −0.00051 to 0.06132. The signal-to-noise and elasticity-wedge legs are power-gated, leaving the Holm family incomplete.

The correct conclusion is weak, concentrated quote synchronization and no promoted HMP claim. We do not identify collusion, communication, provider algorithm, or intent. This is stronger science than treating every correlation as evidence: the property chain specifies exactly what future observations must add.

5.1 A causal simulator result with a failed transport screen

The simulator pairs provider learners while preserving every marginal reward sequence and shuffling only common signal order. This makes the intervention causal *inside the simulator*. In a focal two-UCB-agent prisoner's-dilemma environment, coupled signals increase innovation correlation, all-high play, and mean buyer price. The one-seed screen gives a mean buyer-price effect of +0.12975, positive in 72.2% of focal cells. Across epsilon-greedy and static mixtures, however, the effect is +0.00343 and positive in 52.8% of cells. The executable verdict is `mechanism_validated=false`.

This negative transport result is substantively useful. Signal coupling is not a generic consequence of dynamic pricing; it depends on how agents map common noise into exploration. Any theory claiming universal collusion from public signals is too coarse for this market. The simulation instead motivates a critical-memory question: which learner memory and signal-to-noise combinations make correlated exploration statistically discoverable in finite time? That is a separate learning-theory problem, not asserted here.

6 An Auditable Router Design

6.1 Score, commitments, and exploration

Theorems 1–3 suggest a concrete design.

(1) Compute a generalized score like (8), with task-conditional latency, failure, and quality predictions estimated out of sample.
(2) Replace advertised capacity with an execution-contingent commitment: finite slots, collateral, or SLA credits. Publish how retries enter score and payment.
(3) Use bounded exploration so new or small providers can generate evidence; cap concentration and common-mode exposure.
(4) Charge the router through a fee separated from selected spend, while reporting both user generalized cost and router revenue.

The score need not collapse every objective into one undocumented number. The router can publish a Pareto frontier over user generalized cost, router revenue, provider viability, concentration, and completion. Different user classes can choose points on that frontier, provided the accounting is explicit.

6.2 Robust policy optimization

Let θ parameterize score coefficients, exploration, caps, and fee design. Let Ω be a declared set of cost, quality-value, capacity, and provider- learning models consistent with the evidence. One implementable objective is

$$\min_{\theta} \sup_{\omega\in\Omega} \mathcal{R}_U(\theta;\omega) + \lambda_P \max_{i} \mathcal{E}_i(\theta;\omega) + \lambda_C \max_{S:|S|\le k} \mathcal{E}_S(\theta;\omega) + \lambda_V \mathcal{V}(\theta;\omega), \tag{12}$$

subject to completion, concentration, budget, and capacity constraints. $\mathcal{R}_U$ is user regret relative to the best feasible full-information policy; $\mathcal{E}_i$ and $\mathcal{E}_S$ are unilateral and coalition exploitability; and $\mathcal{V}$ collects constraint violations. This is not a claim that the router knows welfare. It is a way to report which uncertainties drive a design recommendation.

6.3 Historical replay

On 14 observed days containing 18,947 hourly menus, 71 models, and 48 providers, a training-only selection chooses price exponent 1.25 and exploration 0.10. In the post-16-July holdout, the rule reduces the quote-coupling proxy by 49.3%, increases displayed price by 2.08%, and changes public reliability by +0.0059 percentage points. The replay holds menus and provider behavior fixed. It therefore measures a mechanical frontier, not equilibrium, revenue, or welfare. It is a candidate arm for a randomized policy experiment.

6.4 The policy-frontier experiment

The replay suggests a prospective experiment whose estimands are narrower than "welfare" but decision-relevant. Randomize request triplets within model–prompt–time blocks among (i) the production delegated policy, (ii) a price-ordered policy with fallback, and (iii) the adaptive generalized score. Freeze the adaptive score before opening outcomes. The primary vector is attempted-request spend, completion, total latency including retries, and a blinded task-fidelity score. Provider choice, fallback position, and public enforcement state are mechanism variables rather than outcomes to condition on.

Because routing treatments can change congestion, inference should randomize time clusters or use switchbacks after a low-load pilot, report assignment-first intention-to-treat effects, and preserve bounded-outcome randomization intervals. No scalar index should be primary unless the user-value weights are frozen in advance. Instead, report the joint Pareto set and the range of value weights for which each policy is preferred. A second, provider-facing phase randomizes execution-contingent capacity commitments among willing endpoints and measures admission, delivery, and post-experiment retention. This separates the short-run routing effect from the longer-run entry and investment response that a menu replay cannot recover.

The experiment has three promotion gates. First, the adaptive arm must preserve randomization integrity and overlap across models, hours, and provider sets. Second, its completion and fidelity lower bounds must exceed declared safety floors. Third, any welfare statement must remain valid over a preregistered box of user values and provider cost bounds. Failure of the third gate still permits a useful multi-outcome mechanism comparison; it only blocks scalar welfare.

6.5 Practical provider types

The mechanism must work across three economic technologies even though public price regimes do not identify them.

- *Owned or reserved capacity*: high fixed cost, low short-run marginal cost, and a strong incentive to maintain utilization. Capacity commitments are credible when backed by sunk infrastructure but still require outage penalties.
- *Spot-dependent providers*: flexible entry and state-dependent cost. They provide resilience but may disappear exactly during common scarcity. State-contingent commitment prices are more appropriate than a fixed share.
- *Model authors and anchor adopters*: an author quote can coordinate expectations, while identical third-party quotes may reflect default menus rather than common costs. The router should not treat an anchor as a cost certificate.

Free entry is not automatically efficient. An entrant adds a redundant delivery path but diverts requests and may increase retry or monitoring costs, echoing the classic business-stealing result [11]. Capacity-certified entry is preferable to counting endpoint labels: it rewards incremental delivered capacity rather than duplicated menus.

7 Related Work

Multi-model routers such as FrugalGPT, RouteLLM, and RouterBench optimize across models [4, 8, 13]. We hold the model label fixed and study provider procurement; the two routing layers compose. Multidimensional procurement shows why price and quality must be scored jointly [3]; two-sided platform theory clarifies fee and participation externalities [14].

Brown and MacKay study heterogeneous pricing technology and repricing frequency [2]. Their framework motivates our duration and regime transition tests, not a provider-type label. Hansen, Misra, and Pai show that independent, misspecified pricing learners can produce supra-competitive prices through correlated experiments [6]; our ordered property chain and signal-order intervention isolate that mechanism and currently fail to transport it. Johnson, Rhodes, and Wildenbeest show how platform prominence can change algorithmic competition [9]. Inference routing adds hidden eligibility, fallback, and stochastic delivered quality.

The closest empirical benchmark uses OpenRouter and Azure data to study the emerging market for intelligence [5]. We focus on provider- level microstructure and the gap between displayed and delivered liquidity. Switchback and marketplace experiments face interference [1, 7]; our assignment-first probes and first-position estimands are designed around that problem. Price focal points and the reflection problem motivate hard nulls for anchor following rather than causal stories from ties alone [10, 12].

8 Limitations and Conclusion

The public panel is measured in weeks, not quarters. Owned probes cover one account, region, workload, and budget. Capacity, provider cost, user task value, private rebates, and market-wide flow are not observed. Public enforcement fields are router-published aggregates rather than request logs. The current quality panel is selected and incomplete. Historical replays hold menus fixed; simulator interventions are causal only inside the simulator. We therefore do not estimate dumping, literal front-running, communication, collusion, or a scalar welfare loss.

Within those limits, the market description changes sharply. Open-weight execution is not a transparent commodity exchange. It is a two-layer procurement market in which public prices sit above hidden admission, scoring, and fallback. The empirical evidence rejects several tempting simplifications: inverse-square simulated share is not realized flow; active undercutting is not a stable provider type; current response timing does not support a simple reactive collusion story; and premium prices are not yet explained by observed QoS.

The design response is not to declare a better exponent. It is to expose the missing contracting dimensions. Generalized scoring has an exact regret interpretation, approximate scores have a robust guarantee, and hidden capacity creates a formal impossibility for quote-only routing. Execution-contingent prices, capacity commitments, verified quality, bounded exploration, and fee separation turn those results into a testable mechanism. The next empirical step is a randomized policy frontier with delivered outcomes and declared value bands; the next data step is provider- or router-side capacity and attempt logs. Until then, the honest output is a set of feasible mechanisms, not one estimated welfare optimum.

A Proofs

Conditional share elasticity and markup. Write $W_{-i} = \sum_{j \neq i} p_j^{-a}$. Direct differentiation gives

$$\frac{\partial s_i}{\partial p_i} = -\frac{a}{p_i} s_i(1 - s_i).$$

The first-order condition for $(p_i - c_i)s_i$ is $1 - a(p_i - c_i)(1 - s_i)/p_i = 0$, yielding the Lerner expression. At a symmetric profile $s_i = 1/n$; solving for p gives the formula. A finite positive solution requires $a(n-1) > n$. □

Proof of Theorem 1. The Gibbs minimizer satisfies $x_i^g = e^{-\eta g_i}/Z_g$, so $g_i = -\eta^{-1} \log x_i^g - \eta^{-1} \log Z_g$. Substitution into (9) gives

$$F_g(x) = \eta^{-1} \sum_i x_i \log \frac{x_i}{x_i^g} - \eta^{-1} \log Z_g.$$

The second term equals $F_g(x^g)$, proving (10). □

Proof of Theorem 2. For every $x \in \Delta_n$, $|F_g(x) - F_{\widehat{g}}(x)| = |\langle x, g - \widehat{g} \rangle| \leq \epsilon$. Optimality of $x^{\widehat{g}}$ for $F_{\widehat{g}}$ therefore implies

$$F_g(x^{\widehat{g}}) \leq F_{\widehat{g}}(x^{\widehat{g}}) + \epsilon \leq F_{\widehat{g}}(x^g) + \epsilon \leq F_g(x^g) + 2\epsilon.$$

The lower bound follows from optimality of x^g. □

Proof of Theorem 3. A quote-only router must use the same allocation in both observationally equivalent economies. Let x be the probability assigned to provider 1. Its regret is $\Delta(1 - x)$ in economy A and Δx in economy B; the larger is at least $\Delta/2$. If $x \in \{0, 1\}$, the larger is Δ. □

B Evidence and Claim Ledger

The claim ledger is as follows. Pricing regimes use 365 frozen training pairs and 360 holdout pairs but do not identify provider cost or intent. The active- response screen uses 1,006 paired timing events and a cluster interval but is not randomized treatment. The inverse-square restriction is rejected as realized market-wide allocation. H81 identifies a bounded, narrow fallback decomposition, not a general router policy or market share. Residual QoS tests and zero valid utilization cells do not identify technology rent, dumping, or marginal cost. The concentrated WF18 screen with $p = 0.110$ does not identify collusion, communication, or a provider algorithm. The adaptive replay does not identify welfare or equilibrium, and the simulator intervention does not establish live- market causality.

The welfare audit evaluates ten necessary decentralization conditions: five are not identified, three are approximately consistent, one is inconsistent because retry externalities are not internalized, and one is power-gated. This means the full welfare conjecture is not licensed; it is not an estimate that welfare is low. The readiness ledger has one ready object, three power-gated objects, and eight uncollected objects.

C Reproducibility

The empirical figure is generated by the checked-in figure-build script from two aggregate JSON files. WF16, WF18, the adaptive replay, and the signal-coupling simulator are available through the repository's analysis and market-environment entry points. Remote workflows freeze one Hugging Face revision, hydrate it once, run assignment-only preflights, remove owned request rows, and publish aggregate artifacts. Unit tests cover residualization, circular-shift inference, multiplicity, claim flags, signal shuffling, and heterogeneous learner screens. The source revision and protocol hash are stored inside every summary.

References

[1] Iavor Bojinov, David Simchi-Levi, and Jinglong Zhao. 2023. Design and Analysis of Switchback Experiments. *Management Science* 69, 7 (2023), 3759–3777. doi:10.1287/mnsc.2022.4583

[2] Zach Y. Brown and Alexander MacKay. 2023. Competition in Pricing Algorithms. *American Economic Journal: Microeconomics* 15, 2 (2023), 109–156. doi:10.1257/mic.20210158

[3] Yeon-Koo Che. 1993. Design Competition Through Multidimensional Auctions. *The RAND Journal of Economics* 24, 4 (1993), 668–680. doi:10.2307/2555752

[4] Lingjiao Chen, Matei Zaharia, and James Zou. 2023. FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance. *arXiv preprint arXiv:2305.05176* (2023). https://arxiv.org/abs/2305.05176

[5] Mert Demirer, Andrey Fradkin, Nadav Tadelis, and Sida Peng. 2025. *The Emerging Market for Intelligence: Pricing, Supply, and Demand for LLMs.* Working Paper 34608. National Bureau of Economic Research. doi:10.3386/w34608

[6] Karsten T. Hansen, Kanishka Misra, and Mallesh M. Pai. 2021. Frontiers: Algorithmic Collusion: Supra-competitive Prices via Independent Algorithms. *Marketing Science* 40, 1 (2021), 1–12. doi:10.1287/mksc.2020.1276

[7] David Holtz, Felipe Lobel, Ruben Lobel, Inessa Liskovich, and Sinan Aral. 2025. Reducing Interference Bias in Online Marketplace Experiments Using Cluster Randomization: Evidence from a Pricing Meta-Experiment on Airbnb. *Management Science* 71, 1 (2025), 390–406. doi:10.1287/mnsc.2020.01157

[8] Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt Keutzer, and Shriyash Kaustubh Upadhyay. 2024. RouterBench: A Benchmark for Multi-LLM Routing System. *arXiv preprint arXiv:2403.12031* (2024). https://arxiv.org/abs/2403.12031

[9] Justin P. Johnson, Andrew Rhodes, and Matthijs Wildenbeest. 2023. Platform Design When Sellers Use Pricing Algorithms. *Econometrica* 91, 5 (2023), 1841–1879. doi:10.3982/ECTA19978

[10] Christopher R. Knittel and Victor Stango. 2003. Price Ceilings as Focal Points for Tacit Collusion: Evidence from Credit Cards. *American Economic Review* 93, 5 (2003), 1703–1729. doi:10.1257/000282803322655509

[11] N. Gregory Mankiw and Michael D. Whinston. 1986. Free Entry and Social Inefficiency. *The RAND Journal of Economics* 17, 1 (1986), 48–58.

[12] Charles F. Manski. 1993. Identification of Endogenous Social Effects: The Reflection Problem. *The Review of Economic Studies* 60, 3 (1993), 531–542. doi:10.2307/2298123

[13] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M Waleed Kadous, and Ion Stoica. 2024. RouteLLM: Learning to Route LLMs with Preference Data. *arXiv preprint arXiv:2406.18665* (2024). https://arxiv.org/abs/2406.18665

[14] Jean-Charles Rochet and Jean Tirole. 2003. Platform Competition in Two-Sided Markets. *Journal of the European Economic Association* 1, 4 (2003), 990–1029. doi:10.1162/154247603322493212